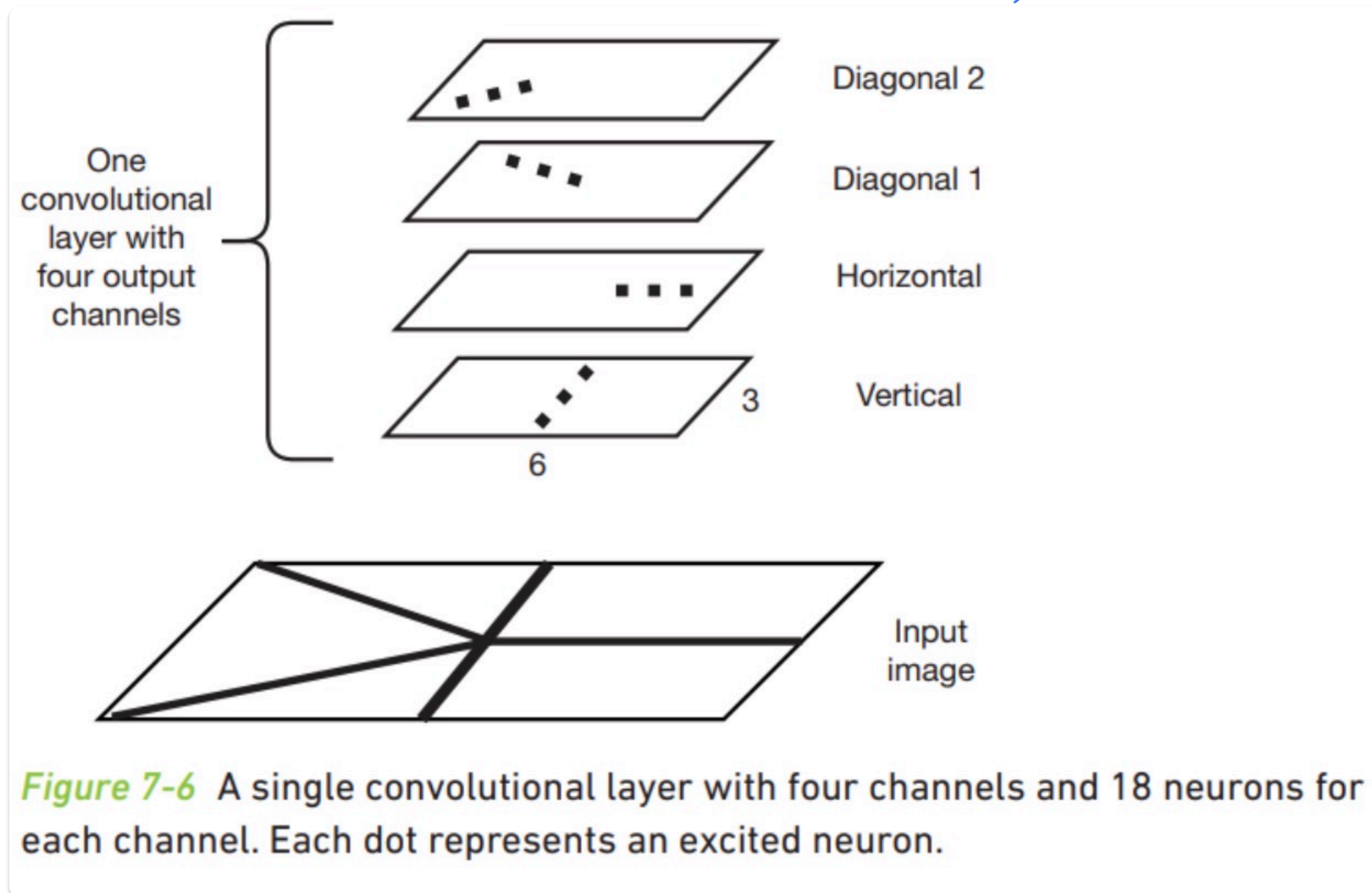


1. A Single Convolution with Multiple Output Channels:

- A. a convolutional layer with four feature maps, and each single feature map identifies a specific feature in the image.
- the bottom-most output channel can identify vertical lines.
 - the next output channel can identify horizontal lines.
 - the two top output channels can identify diagonal lines.
- each feature map consists of 3×6 neurons indicated by the numbers 3 and 6 in the figure.

B.



2. Dimensional Consistency within Feature Maps:

every feature map within the same convolutional layer has an identical num of neurons.

this means that all feature maps in a single convolutional layer share the exact same Width and Height.

3. Num of Neurons in a Convolutional Layer:

$$N_c = (\text{Width} \times \text{Height}) \times C_{\text{out}}, \left[\begin{array}{l} N_c = \text{num of neurons in a convolutional layer.} \\ \text{Width} \times \text{Height} = \text{the grid dimensions of the neurons arranged within a single feature map in a convolutional layer} \\ C_{\text{out}} = \text{num of output channels in a convolutional layer.} \end{array} \right]$$